Roberto Piva

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Senior Software Engineer with 12 years of experience, 10 of them dedicated to PEC, Italy's legally binding, eIDAS-compliant registered email (3M+ accounts, second-largest provider). Java backends, Kubernetes and Azure, cloud cost optimization (FinOps), and AI-agent automation of engineering workflows.

Experience

Mar. 2026 – Current
Senior Software Engineer: Registered email, *InfoCert S.p.A.*, Padua – Italy.
- Modernizing a central Java component of the webmail backend (~2M users) by re-architecting it for maintainability and scalability with cloud cost in mind, and planning a zero-downtime migration
- Took ownership of a Go codebase previously maintained by consultants: lead its design and code review, and ship its production changes, implemented with AI coding agents
- Automated the team's release and incident workflows: an Agent Skill, adopted by the team, that cut release due diligence from over an hour to minutes, and a Python tool that saves ~2 hours a week by updating dozens of related tickets at once

Jan. 2025 – Feb. 2026
Senior Software Engineer, FinOps: Registered email, *InfoCert S.p.A.*, Padua – Italy.
Cut the team's multi-million-dollar cloud bill by 10% without service impact
- Analyzed resource usage (compute, storage, managed services) and usage patterns with scripts combining inventory and monitoring data
- Defined, guided and carried out cost cuts: VM resizing, automated storage tiering, unused resource shutdown, off-hours scale-down, live Kubernetes node-type switch
- Installed KEDA on AKS for queue-based autoscaling, aligning cost with load
- Led a storage access cost reduction on a critical service end to end: analysis, release plan, stakeholder coordination and zero-downtime deployment
- Carried out Terraform updates of critical infrastructure (Kubernetes, databases), coordinating with Microsoft support and consultants

2016 – 2024
Senior Software Engineer: Registered email, *InfoCert S.p.A.*, Padua – Italy.
- Eased the on-call workload by creating Grafana alerts, dashboards and runbooks
- Centralized the team's logs and metrics in Loki and Mimir by deploying shared Grafana Alloy agents, along with security agents
- Moved ~20 services across 2 sub-teams to Azure by coordinating the migration:
  - Gathered common requirements from stakeholders
  - Defined a standard Helm-based Azure DevOps pipeline and VM-to-container guidelines
  - Guided the sub-teams through migrating their workloads
- Helped win and keep the CONSIP public administration PEC contract by building the Java platform that onboarded tens of agencies and hundreds of thousands of accounts
- Released the team's first container project (2020), a Spring application on AWS: EKS, ECR, S3, SQS and RDS PostgreSQL
- Grew the team's new hires from junior to proficient through mentoring, and onboarded the team's consultants

2014 – 2016
Software Engineer: Registered email, Digital signature, Digital long-term archiving, *InfoCert S.p.A.*, Padua – Italy.
- Designed and developed backends for customer-facing web applications
- Evolved business-critical legacy systems without regressions, adding tests before changing untested code
- Raised code quality across all teams by introducing code review, static analysis and best practices (SOLID, testing, dependency injection), through internal talks and mentoring
- Sped up development by building internal Java libraries that remove boilerplate

Sep. 2013 – Mar. 2014
Software Developer (Internship): Windows 8 touchscreen application for fashion retail salesmen, *Maxima S.R.L.*, Padua – Italy.
- Designed a new Windows 8 application from scratch
- Learned C#, XAML design and asynchronous programming

Technical skills

Methods and tools: **FinOps, AI agents, Agent Skills, System design, API design, Testing, Maven, Dependency Injection**.

Programming languages: **Java, SQL, Bash, Python, Go**.

Technologies: **Linux, Kubernetes, Helm, KEDA, Terraform, Azure, Azure DevOps, Jenkins, AWS, Grafana (Prometheus, Loki, Mimir, Alloy), PostgreSQL, Claude Code, Jakarta EE, Quarkus, Spring, REST services**.

Publications and products

2017 – 2018 **Viper – A generator and a framework for injecting configurations via CDI**, *personal project*, GitHub civitz/viper.
Developed a library to generate a configuration injector and validator for CDI environments.

2016 – 2017 **NotSoJug**, *Java User Group*, GitHub notsojug/jug-material.
Presentations for colleagues and new hires on coding and architecture design patterns and on new libraries, each with code examples published on GitHub.

November 2013 **MIREX 2013**, *Combining timbric and rhythmic features for semantic music tagging.* (Orio, N. and Piva, R.).
Developed an algorithm to infer semantic tags for music pieces from a set of annotated songs. Modeled the relationship between audio features and tags in a Hidden Markov Model-inspired structure.

Education

March 2013 **Master's Degree – Computer Science Engineering**, *University of Padua*, 110/110 summa cum laude.
Thesis: *Combining timbric and rhythmic features for semantic music tagging*. Supervisor: Prof. Nicola Orio.

November 2010 **Bachelor's Degree – Computer Science Engineering**, *University of Padua*, 110/110.

July 2007 **Technician – Informatics**, *Istituto Tecnico Industriale Statale J.F. Kennedy*, Pordenone – Italy.

Other Experiences

August 2011 – June 2012 **Algorithm enhancements and benchmarking for a high-performance external memory sorting library**, *Psort project*, University of Padua.

2011 – 2012 **Design and implementation of a GUI plugin**, *PariPari project*, University of Padua, a peer-to-peer, plugin-based distributed platform with instant messaging features, written in Java.

2009 – 2010 **Implementation of XMPP support for an instant messaging plugin**, *PariPari project*, University of Padua.

Languages

Italian: Mother tongue

English: C1[1]

Interests

Work Related: I am interested in system design, artificial intelligence and code quality. I am a techie and like to stay up to date on every tech-related subject. I run a homelab for my self-hosted applications, monitored with Grafana, Prometheus, Loki and Alloy, alerting included.

Other: I am fascinated by privacy threats and mass communication. I enjoy listening to music of all kinds, especially when deep-diving into a problem, and movies too. I am a passionate home cook, and I love to travel.

[1] According to Europass CEFR self-evaluation reference, `https://europass.europa.eu`